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Code

Immigration Enforcement Across U.S. Borders

Sydney Whalen

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Data Summary

The data set contains records of US immigration encounters across the Northern Border, Southwest Border, and encounters that occur in the US states outside of these border regions. Encounters is defined as individuals who were stopped by customs or border security and rejected entry. The basis for rejection are outlined under immigration laws title 8 and title 42. Title 8 gives authority to facilitate an individuals removal without a standard immigration hearing because the individual can't give proper legal reasoning for staying in the U.s (such as asylum). Title 42 was enacted during COVID to reject entry based on public health concerns. The data details these encounters from 2020 to 2023, including demographic information about the individuals encountered as well as information about what protocols and regions handled the encounter.

With significant emphasis placed on Southwest border security during Trumps first term, the goal of this data exploration is to identify how immigration fear mongering along the Southwest border influences customs protocol and migration patterns compared to other border regions. Although this data represents years where Trump did not serve as president, its important to analyze the lasting effects of immigration policy and social perceptions of immigration.

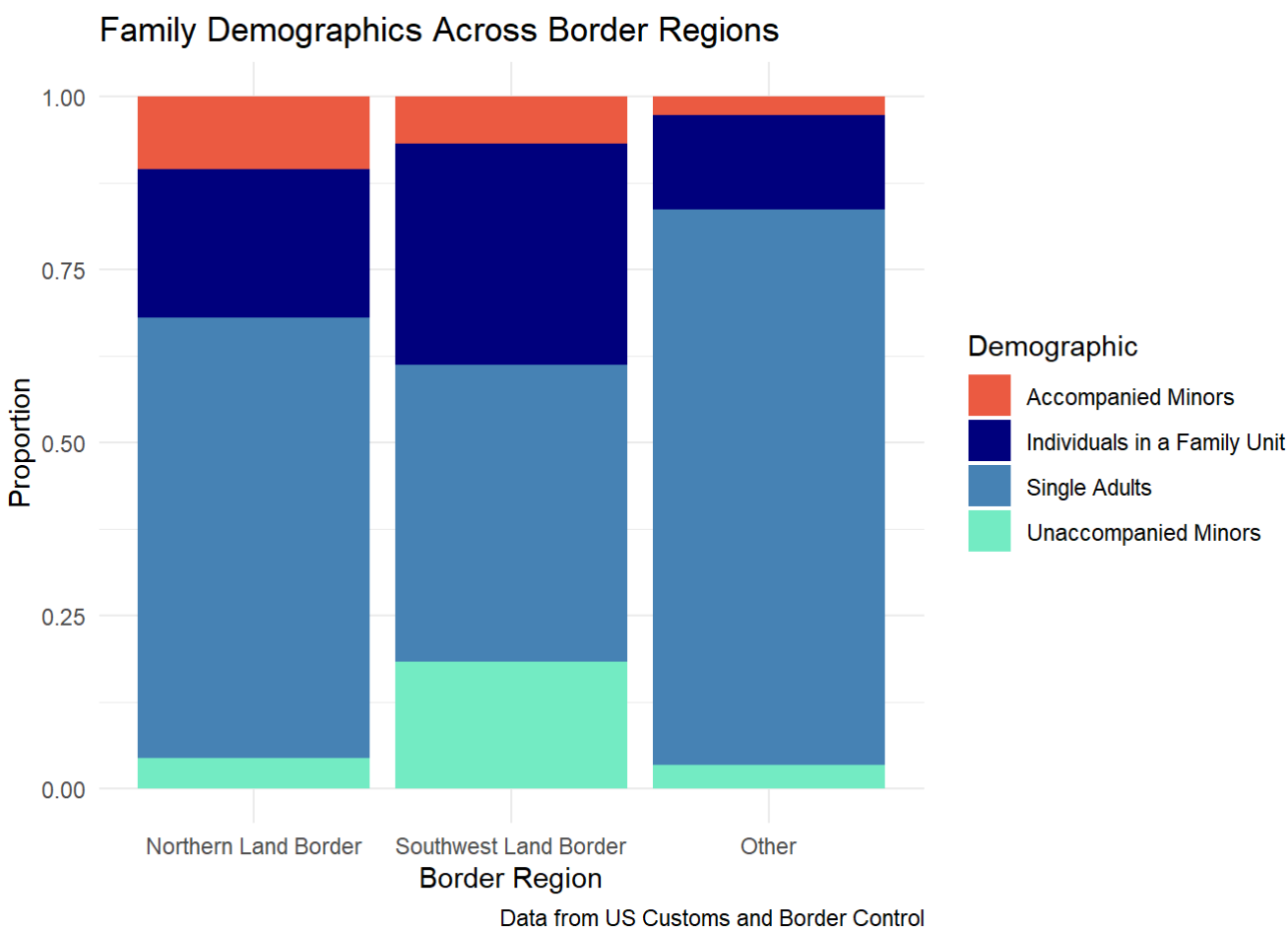
The data is provided by US Customs and Border Control. (https://www.cbp.gov/document/stats/nationwide-encounters (https://www.cbp.gov/document/stats/nationwide-encounters))

Originally accessed through tidytuesday. (https://github.com/rfordatascience/tidytuesday/blob/main/data/2024/2024-11-26/readme.md (https://github.com/rfordatascience/tidytuesday/blob/main/data/2024/2024-11-26/readme.md))

Whose Crossing?

FIGURE 1

```
ggplot(cbp_fy20_23_state,aes(x=`Land Border Region`))+
  geom_bar(aes(fill=Demographic),position="fill")+
  scale_fill_manual(values=c("tomato2","navy", "steelblue","aquamarine2"))+
  theme_minimal()+
  labs(x="Border Region", y="Proportion", title="Family Demographics Across Border R
egions",
      caption="Data from US Customs and Border Control")
```



Analysis: Figure 1 displays the proportions of each demographic found to be crossing in each border region. The Southwest border see's the highest proportion of individuals in a family unit and unaccompanied minors as well as the smallest proportion of single adults crossing. The other category (encounters taking place within the US beyond border entries) sees the highest proportion of single adults. In analyzing these proportions its worth questioning why the border with the highest proportion

of individuals living in family units or individuals who are underage, is also the border that is militarized the most. Comparatively the other border regions see a higher proportion of single adults, the demographic most frequently associated with criminal activity.

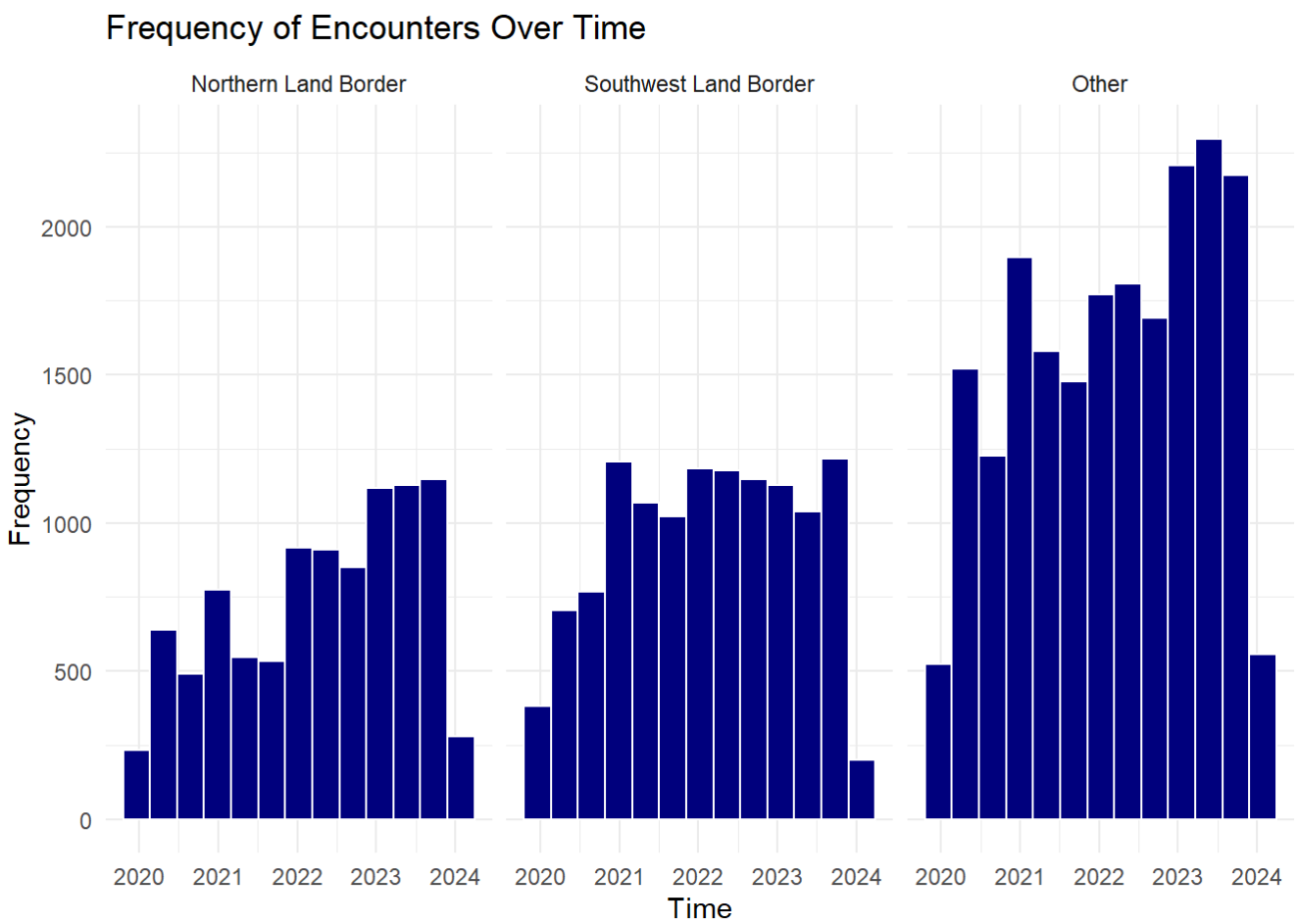
I chose a proportioned bar chart to represent demographic data because in mapping region to the x axis, it is very easy to compare the proportions to each other since they are displayed side by side. Representing this information through color is important to create distinct visual proportions that can be easily differentiated from each other.

Comparing Migration Trends

Growth Trends

FIGURE 2

```
ggplot(cbp_fy20_23_state,aes(x=`Month Year`))+
  geom_histogram(color="white",fill="navy", binwidth=125)+
  theme_minimal()+
  labs(x="Time", y="Frequency", title="Frequency of Encounters Over Time")+
  facet_wrap(~`Land Border Region`)
```



Analysis: Figure 2 shows a clear upward trend in migration over all border regions from 2020 to the end of 2023*. This corroborates frequent reports of an increase in immigration (https://www.cbsnews.com/news/what-is-title-8-immigration-law-vs-title-42-border-policy/ (https://www.cbsnews.com/news/what-is-title-8-immigration-law-vs-title-42-border-policy/)). These reports typically focus on Southwest border crossings since special political emphasis has been placed on that territory, however the data illustrates rises in immigration from all regions.

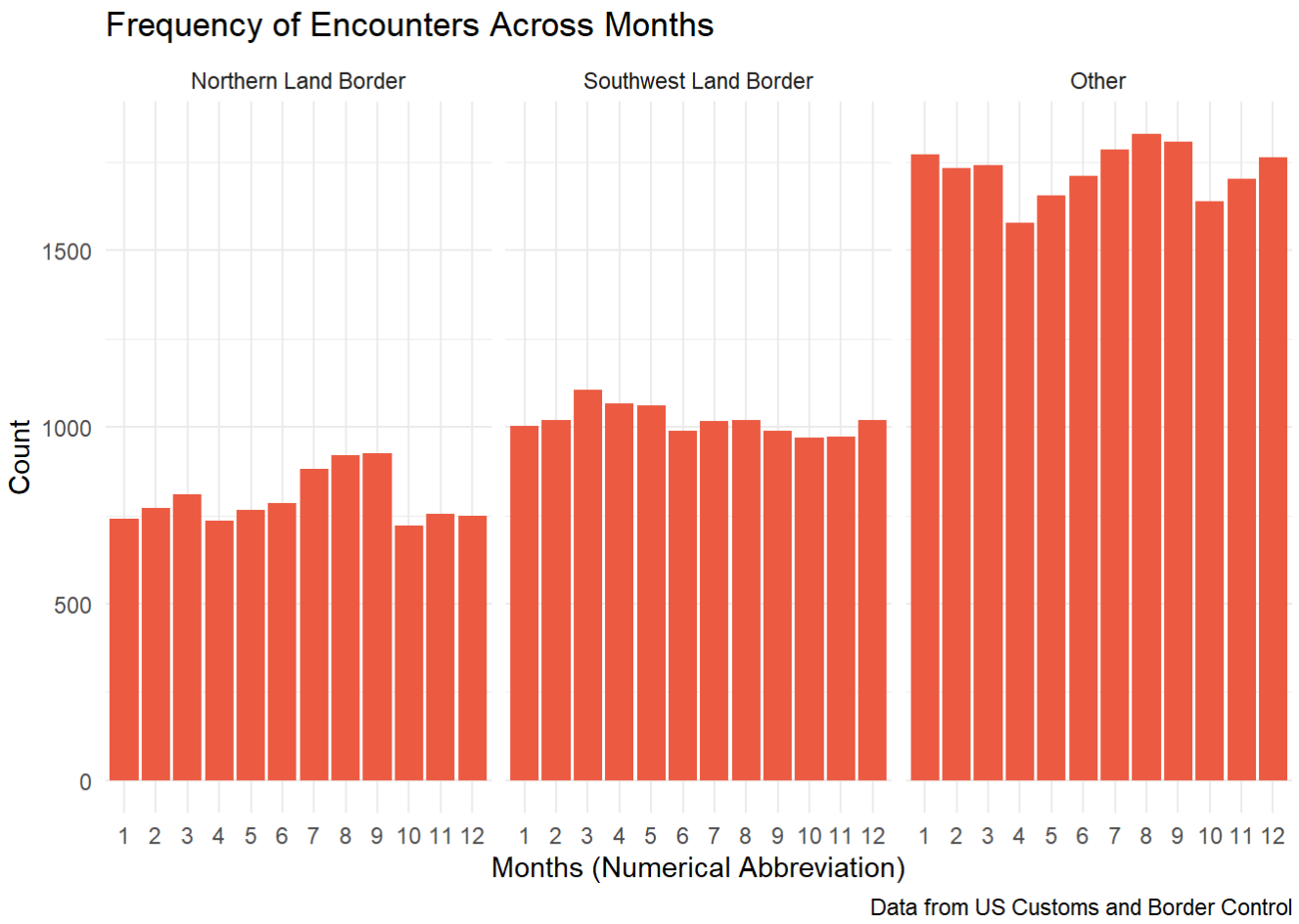
I chose a histogram to represent the distribution because I am mapping a continuous time variable to counts of frequency. The shape this data curates is a helpful visual cue in analyzing trends, specifically overall growth and specific peaks.

*Note: 2024 is not a fiscal year provided in the data set. It is listed as an axis mark to represent the end boundary for the data.

Seasonal Trends

FIGURE 3

```
ggplot(cbp_fy20_23_state,aes(x=`Month (abbv)`))+
  geom_bar(fill="tomato2")+
  scale_x_discrete(labels=c("1","2","3","4","5","6","7","8","9","10","11","12"))+
  theme_minimal()+
  labs(x="Months (Numerical Abbreviation)",y="Count", title="Frequency of Encounters
Across Months", caption="Data from US Customs and Border Control")+
  facet_wrap(~`Land Border Region`)
```



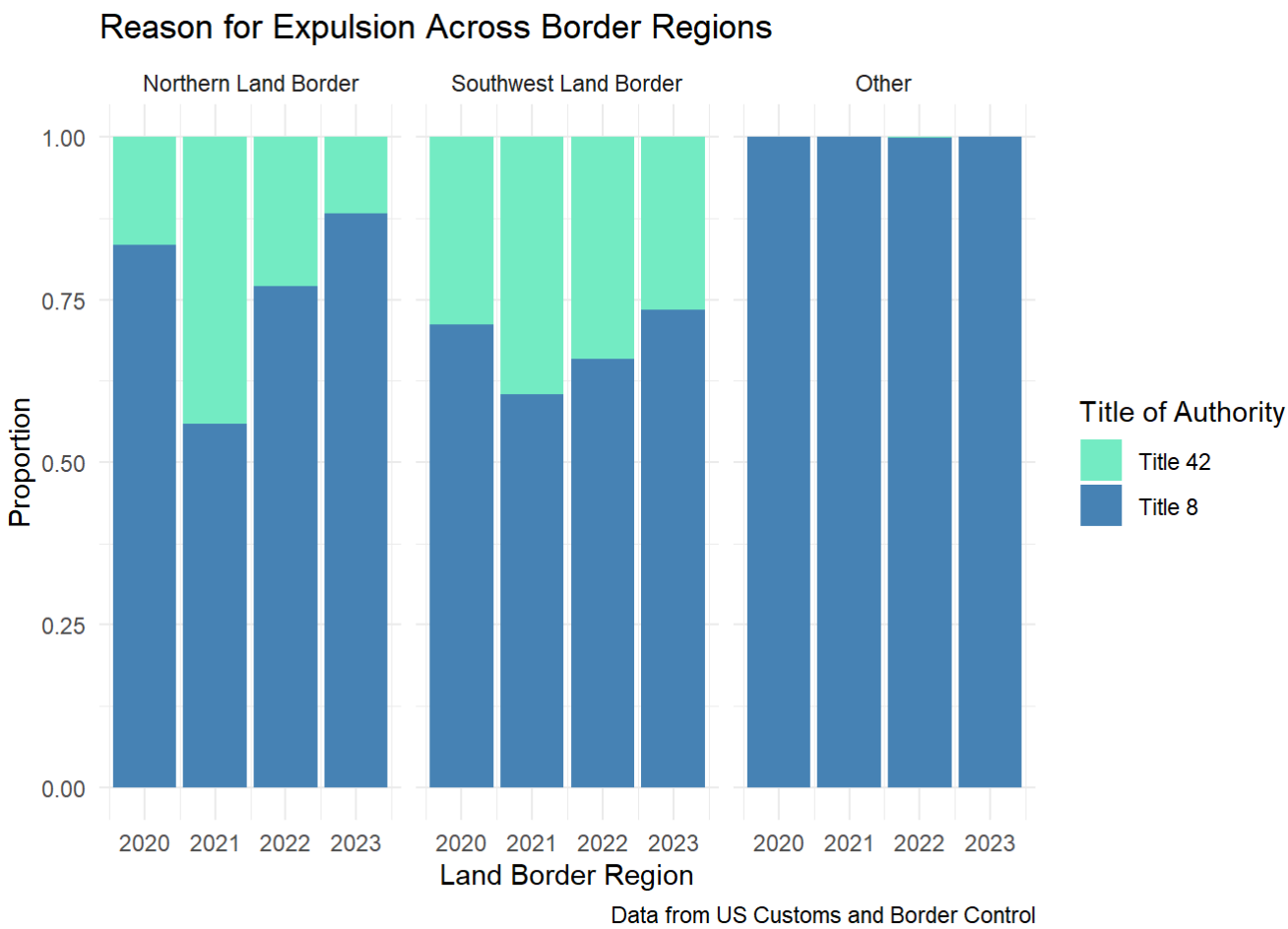
Analysis: Despite a clear upwards trend in border encounters over the years, when analyzing if migration trends have any seasonal patterns its clear the encounters that occur in each border region stay relatively uniform through the year. Notice the subtle peak in the summer months for the Northern Land Border, there is room to speculate that warmer weather conditions in the North may lead to a rise in border crossings. In addressing the uniformity of these frequency distributions, a follow up question might be if its reasonable to assume no natural seasonal migration patterns exist, or if the uniformity can be attributed to how border patrol processes these encounters.

I chose a bar chart to represent the distribution because I am mapping a categorical variable (months) to counts of frequency. The shape this data curates (or lack thereof) is a helpful visual cue in analyzing trends.

Comparing Enforcement Protocals

FIGURE 4

```
ggplot(cbp_fy20_23_state,aes(x=`Fiscal Year`))+
  geom_bar(aes(fill=`Title of Authority`),position="fill")+
  theme_minimal()+
  scale_fill_manual(values=c("aquamarine2","steelblue"))+
  facet_wrap(~`Land Border Region`)+
  labs(x="Land Border Region",y="Proportion", title="Reason for Expulsion Across Bor
der Regions",
       caption="Data from US Customs and Border Control")
```



Analysis: Figure 4 shows the proportion of expulsions using title 8 versus title 42 for each region. Noticeably the other region (encounters taking place within the US beyond border entries) has no expulsions due to title 42, which is reasonable considering these encounters are clearly illegal entries ignoring outlined due process and immediately falls under general immigration law. In comparing the North and Southwest borders, the Northern border has the highest proportion of title 42 expulsions in 2021, but in general the Southwest employs title 42 in higher proportion over the 202-2023 period. Why would these proportions, specifically in comparing 2022 and 2023, be different? Do the biases of individuals who cross the Southwest border have any relation to how border patrol analyzes these people as a public health threat?

I chose another stacked bar chart to compare proportions through color distinction because the data set is composed of mostly categorical variables. The color choice here is consistent with that used in figure one, and the categories are arranged such that the proportioned data is placed right next to eachother allowing for easiest comparison.